

CLASS 11-CBSE MATHEMATICS

CHAPTER: RELATIONS AND FUNCTIONS-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions [1X3 = 3 MARKS]

1. Given $A = \{1, 2, 3\}$, $B = \{3, 4\}$, $C = \{4, 5, 6\}$, find $(A \times B) \cap (B \times C)$.
2. If $f(x) = (x - a)^2 (x - b)^2$, find $f(a + b)$.
3. If $y = f(x) = \frac{ax-b}{bx-a}$, show that $x = f(y)$.

SECTION B

Q.II Answer the following questions [2X5 = 10 MARKS]

4. Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Write $A \times B$. How many subsets will $A \times B$ have? List them.
5. Let $A = \{1, 2, 3, \dots, 14\}$. Define a relation R from A to A by $R = \{(x, y): 3x - y = 0, \text{ where } x, y \in A\}$. Write down its domain, codomain and range.
6. Find domain of the function $\frac{x^2+2x+1}{x^2-8x+12}$
7. Find the domain and range of real valued function $f(x) = \sqrt{x - 1}$
8. Let $f = \left\{ \left(x, \frac{x^2}{1+x^2} \right) : x \in \mathbb{R} \right\}$ be a function from $\mathbb{R}$ to $\mathbb{R}$. Determine the range of f

SECTION C

Q.III Answer the following questions [3X3 = 9 MARKS]

9. Let $A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$, $C = \{5, 6\}$ and $D = \{5, 6, 7, 8\}$. Verify that:
(i) $A \times C \subset B \times D$
(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
10. Find the domain and range of real valued function $f(x) = \frac{ax+b}{bx-a}$
11. Let $A = \{9, 10, 11, 12, 13\}$ and let $f: A \rightarrow \mathbb{N}$ be defined by $f(n) =$ the highest prime factor of n . Find the range of f

SECTION D



Q.IV Answer the following questions

[4X2 = 8 MARKS]

12. Determine the domain and range of the following relations:

(i) $R = \{(a, b) : a \in \mathbb{N}, a < 5, b = 4\}$

(ii) $S = \{(a, b) : b = |a - 1|, a \in \mathbb{Z} \text{ and } |a| \leq 3\}$

13. If $f(x) = 64x^3 + \frac{1}{x^3}$ and α, β are the roots of $4x + \frac{1}{x} = 3$. Then prove that $f(\alpha) = f(\beta) = -9$

